Cole Snipes

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SUMMARY

Full-stack software engineer and data analyst with a B.S. in Computer Science (May 2026). Builds AI-powered web applications, REST APIs, and machine learning pipelines end to end. Delivered a capstone application now used by a university physical therapy program, across five Agile sprints with a five-person team. Seeking a full-stack or data-focused engineering role.

EDUCATION

University of Indianapolis Graduated May 2026

B.S. Computer Science, concentration in Software Engineering · Indianapolis, IN

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, Rust, R, SQL, HTML/CSS

Frontend: React, Svelte & SvelteKit, Next.js, Vite, Tailwind CSS, WebAssembly

Backend & Data: Node.js, Flask, Django, REST APIs, GraphQL, Firebase (Firestore, Auth, Cloud Functions), SQL/NoSQL

ML & AI: pandas, NumPy, scikit-learn, Keras, LLM API integration (Gemini, Claude, OpenAI), RAG, prompt engineering

Tools & Practices: Git, GitHub Actions, Docker, CI/CD, Playwright, Selenium, Jupyter, Linux, Agile/Scrum, code review

PROJECTS

Curriculum Mapping Application — Krannert School of Physical Therapy 2025–2026

SvelteKit 2, Svelte 5, Firebase, Gemini AI

- Replaced a manual, spreadsheet-based CAPTE process with a web application that maps courses and learning objectives against 93 accreditation standards in one interface.
- Enforced role-based access control for two roles (admin, viewer) at the Firebase Rules layer, not just the UI, so policy could not drift between layers.
- Integrated Gemini AI to surface curriculum alignment gaps, cutting administrator cross-referencing time from hours to minutes per review cycle.
- Delivered across five Agile sprints with a five-person team, validating requirements with the department chair.

Predictive Analysis of Obesity Levels 2024

Python, Keras, scikit-learn, pandas

- Built an end-to-end ML pipeline (cleaning, EDA, feature engineering, training, evaluation) over 2,111 records and 16 features across six obesity classes.
- Compared a neural network (96% accuracy), KNN (95%), and random forest; validated with ROC curves and confusion matrices rather than accuracy alone.
- Identified family history, diet, and physical activity as dominant predictors via chi-square analysis (χ^2=448.9, $p<0.001$).

WebVPython WASM Runner — Glowscript (open source) 2023

JavaScript, WebAssembly, Pyodide, Docker

- Extended an open-source WASM runner so Python physics simulations run entirely in-browser, removing the local install that gated every classroom demo.
- Engineered a secure cross-origin postMessage bridge and modified canvas rendering to enable screenshot export.

n8n-to-Python Workflow Transpiler 2025

Python, AST, CLI

- Built a CLI that converts n8n visual automation workflows into standalone Python via AST-based code generation, removing the n8n runtime dependency for production deployments.
- Chose AST generation over string templating so nested conditionals and dynamic field references survive translation.

EXPERIENCE

Cart Attendant — Meijer, Indianapolis, IN January 2024 – Present

- Maintained cart availability and floor safety across a high-volume store through peak hours with minimal supervision.
- Coordinated with associates on spill response and customer assistance; recognized for reliability.

AWARDS

- Richard Lugar Academic Scholarship
- Presented statistical research to Summer Research Institute faculty; selected for the Scholar Showcase exhibition.